PROPAGATION OF GCH ABOVE A STRONGLY COMPACT CARDINAL

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Editorial note. Complete original mathematical body, both supporting lemmas and embedded bibliography retained. This is the author theorem as stated; mathematical responsibility remains with author, without independent certification, repaired proof or added assumptions.

Editorial note. Original acknowledgement250–253 retained: work used GPT-6-Astra ultra assistance and dynamically invoked GPT-6.1-Sol/GPT-6-Astra; author takes full responsibility. The declaration does not identify an AI-discovered core proof target, and no pure-human or AI-free claim is added.

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Allowed standard prerequisites (editorial).

- Strong compactness, definable elementary embeddings, internal cardinal arithmetic and GCH. Native strong-compactness definition, minimal D covering convention, internal/external cardinality distinction and all proof-support lemmas retained.. The whole original main proof and both essential author lemmas are bundled into one target; local and weaker-critical-point remarks not separate counts.

1. Introduction

Cardinal arithmetic studies the possible values of cardinal exponentiation. Of particular interest is the continuum function $\lambda \mapsto 2^\lambda$. The *Generalized Continuum Hypothesis* (GCH) asserts that this function takes its least possible value at every infinite cardinal: $2^\lambda = \lambda^+$.

Easton's theorem [Kan03, Theorem 10.7] shows that ZFC imposes essentially no constraint on the continuum function $\lambda \mapsto 2^\lambda$ at regular cardinals beyond monotonicity and König's inequality $\mathrm{cf}(2^\lambda) > \lambda$. However, if we assume the existence of suitable large cardinal, at singular cardinals the situation is different: Solovay's classical theorem [Kan03, p. 308] asserts that if κ is strongly compact, then $2^\lambda = \lambda^+$ for every singular strong limit cardinal $\lambda > \kappa$.

Solovay's theorem gives no information about 2^λ for regular $\lambda \geq \kappa$. The natural hypothesis under which one may hope to control these values is GCH below κ: If κ is measurable and GCH holds below κ, then $2^\kappa = \kappa^+$, by a result of Scott [Kan03, Corollary 5.8]; If κ is supercompact and GCH holds below κ, then GCH holds [Kan03, Proposition 22.2]. Woodin [Apt00] asked whether this remains true for strongly compact cardinals, which is weaker than supercompactness.

Question 1 (Woodin, [Apt00])**.** If κ is strongly compact and GCH holds below κ, then does GCH hold?

Apter [Apt00] obtained a negative answer in the absence of the Axiom of Choice. In this paper we answer the question in the affirmative in ZFC.

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Theorem A. *If κ is a strongly compact cardinal and* GCH *holds below κ, then* GCH *holds.*

2. Preliminaries

Our notation and terminology are standard and follow [Kan03]. We work in a transitive model V of ZFC. For an infinite cardinal λ, let $2^{<\lambda} := \sup\{2^\mu \mid \mu < \lambda \text{ is a cardinal}\}$. We say that GCH *holds below* κ iff $2^\mu = \mu^+$ for every infinite cardinal $\mu < \kappa$. For sets X and Y, $X \triangle Y$ denotes their symmetric difference. For an elementary embedding $j : V \to M$, let $\mathrm{crit}(j)$ denote the critical point of j, i.e., the least ordinal moved by j. We write $j[X] := \{j(x) \mid x \in X\}$ for any set X. Superscripts M indicate computations in M; for $X \in M$, $|X|^M$ and $\mathcal{P}(X)^M$ denote the cardinality and power set of X computed in M, respectively. Unless indicated otherwise, cardinalities and power sets are computed in V.

Definition 2.1 (Strong compactness)**.** Suppose that κ is an uncountable cardinal.

(1) For any $\lambda \geq \kappa$, κ is *λ-strongly compact* iff there exists a definable elementary embedding $j : V \to M$ into a transitive class M with critical point κ, such that there exists a $D \in M$ with $j[\lambda] \subseteq D$ and $M \models |D| < j(\kappa)$. In this case, we say that D is *minimal* iff $|C|^M \geq |D|^M$ for every $C \in M$ with $j[\lambda] \subseteq C$.
(2) We say that κ is *strongly compact* iff it is λ-strongly compact for every $\lambda \geq \kappa$.

Notation 2.2. In the following, without loss of generality, we may further assume that $D \subseteq \sup(j[\lambda])$ by taking $D \cap \sup(j[\lambda])$ if necessary.

The next lemma records two consequences of choosing a minimal cover.

Lemma 2.3. *Suppose κ is λ-strongly compact witnessed by an elementary embedding $j : V \to M$ and a minimal D. Let $d := |D|^M$. Then:*

(1) *If $S \in M$ and $|S|^M < d$, then $|\{\alpha < \lambda \mid j(\alpha) \in S\}| < \lambda$.*
(2) *There is a set $R \subseteq d$ cofinal in d with $|R| \leq \lambda$.*

Proof. (1) Let $E := \{\alpha < \lambda \mid j(\alpha) \in S\}$. Suppose for contradiction that $|E| = \lambda$. Fix a bijection $e : \lambda \to E$. By elementarity, $j(e) : j(\lambda) \to j(E)$ is a bijection in M. Let

$$C := \{\beta < j(\lambda) \mid j(e)(\beta) \in S\} \in M.$$

Since $j(e)$ maps C injectively into S, we have $|C|^M \leq |S|^M < d$.

For every $\alpha < \lambda$, we have $e(\alpha) \in E$, so

$$j(e)(j(\alpha)) = j(e(\alpha)) \in S.$$

Thus $j(\alpha) \in C$, and hence $j[\lambda] \subseteq C$. This contradicts the minimality of D.

(2) Fix in M a bijection $f : D \to d$, and put $R := f[j[\lambda]]$. Then $R \subseteq d$ and $|R| = \lambda$. If R were bounded in d, there would be some $\gamma < d$ with $R \subseteq \gamma$. Then $f^{-1}[\gamma] \in M$ would contain $j[\lambda]$ and have M-cardinality $|\gamma|^M < d$, contradicting the minimality of D. Hence R is cofinal in d.

□

3. Proof of the main theorem

The following lemma is the key to the proof of Theorem A.

Lemma 3.1. *Let $\kappa \leq \lambda$ be uncountable cardinals, and suppose that κ is λ-strongly compact, as witnessed by a definable elementary embedding $j : V \to M$ and a*

minimal D. Put $d := |D|^M$ and $F := \{j(A) \cap D \mid A \subseteq \lambda\}$. For every $T \in M$ with $|T|^M \leq d$, we have

$$|F \cap T| \leq 2^{<\lambda}. \tag{1}$$

Proof. We first reduce to the case $|T|^M < d$. By Lemma 2.3(2), fix a set $R \subseteq d$ cofinal in d with $|R| \leq \lambda$. Fix in M an injection $q : T \to d$, and for every $r \in R$ let $T_r := \{t \in T \mid q(t) \leq r\}$. Then $T_r \in M$ and $|T_r|^M \leq |r+1|^M < d$, since d is an infinite cardinal of M. By cofinality of R, $T = \bigcup_{r \in R} T_r$. Thus, if the inequality (1) holds for each T_r, it also holds for T, since $\lambda \cdot 2^{<\lambda} = 2^{<\lambda}$.

We may therefore assume that $|T|^M < d$. Let

$$S := \{\min(X \mathbin{\triangle} Y) \mid X, Y \in T \cap \mathcal{P}(D)^M \text{ and } X \neq Y\}, \qquad E := j[\lambda] \cap S.$$

Since D is a set of ordinals and $T \cap \mathcal{P}(D)^M \in M$, the set S is well-defined and belongs to M. Moreover, $F \cap T \subseteq T \cap \mathcal{P}(D)^M$, and

$$|S|^M \leq |T \times T|^M < d.$$

Since j is injective, Lemma 2.3(1) gives

$$|E| = |\{\alpha < \lambda \mid j(\alpha) \in S\}| < \lambda.$$

Claim 3.1.1. *Suppose that $A, B \subseteq \lambda$ are distinct. Then*

$$\min\big((j(A) \cap D) \mathbin{\triangle} (j(B) \cap D)\big) = j\big(\min(A \mathbin{\triangle} B)\big).$$

Proof. Let $\delta := \min(A \mathbin{\triangle} B)$. By elementarity, $j(A) \mathbin{\triangle} j(B) = j(A \mathbin{\triangle} B)$, and its least element is $j(\delta)$. Now

$$(j(A) \cap D) \mathbin{\triangle} (j(B) \cap D) = \big(j(A) \mathbin{\triangle} j(B)\big) \cap D.$$

This set contains $j(\delta)$, because $j(\delta) \in j[\lambda] \subseteq D$, and it contains no ordinal below $j(\delta)$. Hence its least element is $j(\delta)$. □

Define

$$\Phi : F \cap T \longrightarrow \mathcal{P}(E), \qquad \Phi(X) := X \cap E.$$

We claim that Φ is injective. Take distinct $X, Y \in F \cap T$, and choose $A, B \subseteq \lambda$ such that $X = j(A) \cap D$ and $Y = j(B) \cap D$. Then $A \neq B$. Let $\delta := \min(A \mathbin{\triangle} B)$. By Claim 3.1.1,

$$j(\delta) = \min(X \mathbin{\triangle} Y) \in S.$$

Thus $j(\delta) \in j[\lambda] \cap S = E$, and $j(\delta)$ belongs to exactly one of X and Y. Consequently $j(\delta) \in \Phi(X) \mathbin{\triangle} \Phi(Y)$, so $\Phi(X) \neq \Phi(Y)$. Therefore

$$|F \cap T| \leq |\mathcal{P}(E)| = 2^{|E|} \leq 2^{<\lambda},$$

as required. □

We now prove Theorem A.

Proof of Theorem A. We argue by induction on the infinite cardinals $\lambda \geq \kappa$. Assume that GCH holds below λ. It suffices to prove that $2^\lambda \leq (2^{<\lambda})^+$. Choose a definable elementary embedding $j : V \to M$ and a minimal $D \in M$ witnessing the λ-strong compactness of κ. Let $d := |D|^M < j(\kappa)$.

Since GCH holds below κ, elementarity gives $M \models$ "GCH holds below $j(\kappa)$". Since d is an infinite cardinal of M below $j(\kappa)$, we have $M \models |\mathcal{P}(D)| = d^+$. Let $\theta := (d^+)^M$. Fix in M a bijection $h : \theta \to \mathcal{P}(D)^M$, and put

$$F := \{j(A) \cap D \mid A \subseteq \lambda\}.$$

Since $j[\lambda] \subseteq D$, for every $A \subseteq \lambda$ and $\alpha < \lambda$,

$$\alpha \in A \quad \iff \quad j(\alpha) \in j(A) \cap D.$$

Thus $A \mapsto j(A) \cap D$ is injective, and $|F| = 2^\lambda$.

For every $\beta < \theta$, we have $h[\beta] \in M$ and $|h[\beta]|^M \leq d$, because $\theta = (d^+)^M$. Now applied Lemma 3.1 to $T := h[\beta]$, therefore gives

$$|F \cap h[\beta]| \leq 2^{<\lambda}.$$

Hence $2^\lambda = |F| = |F \cap h[\theta]| \leq (2^{<\lambda})^+$. □

Remark 3.2. The same argument applies locally: if κ is λ-strongly compact and GCH holds below κ, then $2^\mu = \mu^+$ for every infinite cardinal $\mu \leq \lambda$, since λ-strong compactness implies μ-strong compactness whenever $\kappa \leq \mu \leq \lambda$.

Remark 3.3. The proofs above don't use $\mathrm{crit}(j) = \kappa$. If κ is ω_1-strongly compact (in the sense of Definition 2.1, with $\mathrm{crit}(j) \geq \omega_1$ in place of $\mathrm{crit}(j) = \kappa$), the same proofs apply. Consequently, if κ is ω_1-strongly compact and GCH holds below κ, then GCH holds.

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